

VORKUTA, Russian Federation

WMO: 232260

Lat: 67.4919N

Lon: 63.9754E

Elev: 172

StdP: 99.28

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-39.5	-36.4	-42.5	0.1	-39.4	-39.3	0.1	-36.3	17.3	-12.3	15.6	-12.2	2.0	180	0.925

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	25.4	17.3	22.7	15.9	20.1	14.6	18.2	23.4	16.8	21.7	15.3	19.3	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	11.9	20.3	14.7	10.7	19.2	13.2	9.6	17.6	52.4	23.1	47.9	21.9	43.5	19.2	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.3	10.6	DB	-41.0	28.1	3.7	2.1	-43.7	29.6	-45.8	30.9	-47.9	32.1	-50.7	33.6	(4)
(5)			WB	-40.9	20.7	3.4	2.3	-43.3	22.4	-45.3	23.7	-47.2	25.0	-49.6	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.9	-19.2	-19.5	-14.0	-8.5	-1.7	8.0	12.9	9.6	4.5	-3.1	-12.5	-16.1	(6)	
	DBStd	13.08	8.72	8.92	7.93	6.92	5.53	5.75	5.25	3.79	3.76	5.52	7.95	9.29	(7)	
	HDD10.0	5648	905	827	743	556	364	107	29	56	171	406	676	809	(8)	
	HDD18.3	8492	1163	1060	1002	806	621	313	182	272	416	664	926	1068	(9)	
	CDD10.0	214	0	0	0	0	2	46	119	43	5	0	0	0	(10)	
	CDD18.3	17	0	0	0	0	0	2	14	1	0	0	0	0	(11)	
	CDH23.3	175	0	0	0	0	0	27	143	5	0	0	0	0	(12)	
	CDH26.7	30	0	0	0	0	0	4	26	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.1	5.3	5.3	5.4	5.7	5.3	5.0	4.4	4.5	4.6	4.9	5.0	5.6	(14)	
(15) Precipitation	PrecAvg	542	33	32	31	29	38	52	59	69	57	62	41	39	(15)	
	PrecMax	676	58	63	86	58	66	101	102	110	100	107	73	73	(16)	
	PrecMin	367	13	7	7	7	18	20	18	19	3	28	22	19	(17)	
	PrecStd	72	10	15	15	13	13	21	25	28	27	24	11	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.2	-1.6	1.3	5.6	17.9	26.4	29.2	23.7	18.0	9.7	1.2	0.1	(19)	
		MCWB	-1.6	-2.0	-0.1	3.3	11.1	15.9	18.7	18.2	12.9	8.0	0.5	-0.3	(20)	
	2%	DB	-3.8	-3.2	-0.3	3.5	12.5	23.2	27.0	20.5	14.5	6.9	0.1	-1.4	(21)	
		MCWB	-4.3	-3.6	-1.1	1.7	8.4	14.9	18.0	15.3	11.0	5.8	-0.2	-1.9	(22)	
	5%	DB	-5.6	-4.9	-1.5	2.0	8.8	20.5	24.7	18.3	12.3	5.0	-0.7	-2.8	(23)	
		MCWB	-6.0	-5.5	-2.3	0.7	5.7	13.8	17.2	14.3	9.8	4.1	-1.1	-3.4	(24)	
	10%	DB	-7.9	-7.3	-3.2	0.7	5.8	17.7	22.2	16.3	10.3	3.4	-2.4	-4.3	(25)	
		MCWB	-8.5	-7.9	-3.9	-0.1	3.9	12.3	15.9	13.4	8.5	2.7	-2.9	-4.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	-2.0	0.2	3.6	11.8	17.8	20.6	19.6	13.9	8.0	0.7	-0.2	(27)	
		MCDB	-1.2	-1.7	1.0	5.2	16.5	22.9	26.1	22.3	16.1	9.2	0.9	0.0	(28)	
	2%	WB	-4.3	-3.6	-1.0	1.9	8.5	15.8	19.1	16.3	11.7	5.9	-0.1	-1.8	(29)	
		MCDB	-3.9	-3.3	-0.4	3.1	12.3	21.2	24.9	19.1	13.9	6.7	0.1	-1.5	(30)	
	5%	WB	-6.1	-5.4	-2.3	0.8	5.8	14.2	17.8	14.9	10.0	4.2	-1.1	-3.3	(31)	
		MCDB	-5.7	-4.9	-1.6	1.8	8.2	19.7	23.2	17.6	11.7	4.8	-0.7	-2.8	(32)	
	10%	WB	-8.4	-7.8	-3.8	-0.1	3.8	12.6	16.7	13.4	8.6	2.7	-2.9	-4.8	(33)	
		MCDB	-8.0	-7.3	-3.2	0.8	5.9	17.3	21.6	16.1	10.3	3.3	-2.4	-4.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.1	7.5	7.4	6.3	8.3	9.4	7.4	6.1	4.8	6.4	6.7	(35)	
		MCDBR	6.1	6.5	5.7	5.6	10.0	12.8	12.8	11.0	9.4	5.0	5.6	5.9	(36)	
	5% WB	MCWBR	6.1	6.4	5.5	4.7	6.6	6.5	6.2	6.9	6.2	4.2	5.6	5.9	(37)	
		MCWBR	6.2	6.5	5.7	5.6	9.4	12.0	12.0	10.0	8.3	4.8	5.7	6.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.203	0.253	0.299	0.324	0.341	0.351	0.376	0.357	0.315	0.262	0.194	N/A	(40)	
		taud	1.881	2.023	2.059	2.085	2.161	2.400	2.367	2.402	2.470	2.191	1.910	N/A	(41)	
	Ebn at Noon		115	544	721	815	845	852	812	788	740	540	111	0	(42)	
		Edh at Noon	12	53	91	117	122	100	100	87	63	43	10	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.04	0.45	1.96	3.91	5.28	4.95	4.96	3.04	1.73	0.53	0.10	0.00	(44)	
	RadStd		0.01	0.09	0.31	0.35	0.54	0.51	0.74	0.33	0.17	0.09	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+1.83	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	+1.85	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page